Fiscal Year:	FY 2024	**Task Last Updated:**	FY 07/16/2023
PI Name:	Yagoobi, Jamal Ph.D.		
Project Title:	Electrically Driven Liquid Film Boiling: A Two-Phase Heat Transport Device Driven by Electric Conduction Mechanism		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Fluid physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	jyagoobi@wpi.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	508-831-5759
Organization Name:	Worcester Polytechnic Institute		
PI Address 1:	Department of Mechanical Engineering		
PI Address 2:	10 W 32nd St, E1 Building, Room 243		
PI Web Page:			
City:	Worcester	**State:**	MA
Zip Code:	01609-2280	**Congressional District:**	3
Comments:	NOTE PI formerly at Illinois Institute of Technology; moved to Worcester Polytechnic Institute in January 2012. Info per PI (Ed., 1/17/2012)		
Project Type:	Flight	**Solicitation / Funding Source:**	2008 Fluid Physics NNH08ZTT002N
Start Date:	10/01/2016	**End Date:**	09/30/2022
No. of Post Docs:		**No. of PhD Degrees:**	1
No. of PhD Candidates:	2	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA GRC
Contact Monitor:	Hasan, Mohammad	**Contact Phone:**	216-977-7494
Contact Email:	Mohammad.M.Hasan@nasa.gov		
Flight Program:	ISS		
Flight Assignment:	NOTE: End date changed to 9/30/2022 per NSSC information (Ed., 9/22/21)		
Key Personnel Changes/Previous PI:	June 2019 report: No change to report.		
COI Name (Institution):	Didion, Jeffrey M.S. (NASA Goddard Space Flight Center)		
Grant/Contract No.:	NNX16AT09G		
Performance Goal No.:			
Performance Goal Text:			
	NOTE: The NASA Task Book reflects the NSSC project title, "Electrically Driven Liquid Film Boiling: A Two-Phase Transport Device Driven By Electric Conduction Mechanism." The project is also known as "Electrically Driven Liquid Film Flow Boiling - A Two-Phase Heat Transport Device Driven by Electric Conduction Mechanism" (Ed., 7/18/23). ---------- NOTE: Continuation with same Principal Investigator (PI) Jamal Yagoobi, Ph.D. of "Electrically Driven Liquid Film Boiling: A Two-Phase Transport Device Driven By Electric Conduction Mechanism," grant NNX12AR32G. Electrically Driven Liquid Film Flow Boiling - A Two-Phase Heat Transport Device Driven by Electric Conduction		

Task Description:	Mechanism is an International Space Station (ISS) Experiment to fundamentally investigate the effects of gravity on: 1.) electrically generated two-phase flow (liquid/vapor) and 2.) electrically driven liquid film flow boiling. The proposed work is a collaborative effort between the investigators to develop and validate the flight experiment hardware. We propose this effort based upon the successes of our current project: NASA Grant/Cooperative Agreement: NNX12AR32G Electrically Driven Liquid Film Flow Boiling - A Two-Phase Heat Transport Device Driven by Electric Conduction Mechanism. Although the original effort extends through February 2017, we have completed tasks and achieved the technical and programmatic milestones of this project. That is, we have passed all the requirements of the Science Concept Review and received the authorization to advance to the next stage of the experiment development. Therefore, the PI and Co-PI submit this proposal to develop the flight experiment hardware with an effective start date of October 2016. This proposal fundamentally investigates the complex liquid-vapor phase change phenomenon in presence of electrical field and absence of gravity. Thus, this study will provide valuable scientific understanding about competing physical phenomena of flow inertia and bubble dynamics in electrically driven liquid film flow boiling in the presence of electric bubble extraction force (i.e., dielectrophoretic force) in microgravity. This is a proposal to support the development of the experiment flight hardware for the International Space Station, conduct the experiment on–board the International Space Station, support the operations of the experiment and evaluate the experimental data. The proposed period of performance is October 2016 to September 2021. However, this project was extended at no-cost till September 30, 2022. The robust collaboration between Worcester Polytechnic Institute (WPI) and Goddard Space Flight Center (GSFC) can provide both state of the art theoretical investigation and deliver flight qualified experimental hardware; thus, this proposal presents both investigative and hardware development tasks. The PI will maintain overall technical and scientific direction of the experiment development; however, the Co-I will support the flight experimental hardware development in collaboration with Glenn Research Center. Note: This project successfully passed all requirements of the Science Concept Review and has received authorization to advance to the next stage of development. This is to elevate it to the International Space Station flight status as of October 1, 2016, with this continuation grant.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This multi-physics proposal will provide basic scientific understanding about competing physical phenomena of flow inertia and bubble dynamics in electrically driven liquid film flow boiling in the presence of electric bubble extraction force (i.e., dielectrophoretic force) in microgravity. The electrically driven liquid film flow boiling data aboard ISS will determine the critical heat flux, onset of boiling, absence of hysteresis, and bubble characteristics in the absence of gravity. There are a number of potential terrestrial and space applications for this electrohydrodynamic (EHD) driven two-phase heat transport device in all scales (from macro-scale to truly micro-scales). For example, this device can be used to effectively remove heat from electronics in micro- and zero-gravity environment for NASA's future space stations and long term missions. It can also be used in satellites (e.g., embedded system into satellite's multi-functional structure). Terrestrially, it can be used in high intensity electronics cooling as well as in various phase change devices. To achieve these applications, the basic scientific understanding about competing physical phenomena of flow inertia and bubble dynamics in electrically driven liquid film flow boiling in the presence of electric bubble extraction force (i.e., dielectrophoretic force) in microgravity is needed.
Task Progress:	In this project, the electrically driven liquid film flow boiling was experimentally investigated. Specifically, the liquid film, condensing on the surface of condenser, was effectively pumped toward the electric heater with the Electrohydrodynamic (EHD) conduction pumping mechanism. The liquid reaching the electric heater was boiled and the resultant vapor bubbles were effectively removed away from the heater surface by the dielectrophoretic (DEP) force. This resulted in a significant enhancement of boiling heat transfer in the nucleate boiling regime as well as the critical heat flux. The power required to operate the EHD conduction pumping, as well as the DEP mechanism was minimal. Specifically, the experiments conducted in the Multi-Scale Heat Transfer Laboratory at Worcester Polytechnic Institute provided in-depth understanding of the following: a- The effect of EHD conduction pumping on liquid film flow boiling, b- The effect of dielectrophoretic (DEP) force on liquid film flow boiling, c- The combined effect of EHD conduction pumping and DEP force on liquid film flow boiling. The DEP-assisted EHD-driven liquid film flow boiling performance was evaluated on a parabolic flight in November 2019. The results confirmed that EHD and DEP mechanisms raise the critical heat flux, lower heater surface temperature, and successfully sustain boiling in both microgravity and on the ground with low power consumption.
Bibliography Type:	Description: (Last Updated: 01/30/2026)
Abstracts for Journals and Proceedings	O'Connor N, Yagoobi J. "The effect of temperature on the pumping performance of helical electrohydrodynamic conduction pump." 22nd IEEE International Conference on Dielectric Liquids (ICDL), Worcester, Massachusetts, June 25-28, 2023. Abstracts. 22nd IEEE International Conference on Dielectric Liquids (ICDL), Worcester, Massachusetts, June 25-28, 2023. , Jun-2023